NOTES

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Tutorial - 1

1) Use paths to show graphs not isomorphic or find isomorphism.

→

G vertices → $u_1, u_2, u_3, u_4, u_5, u_6, u_7, u_8$

H vertices → $v_1, v_2, v_3, v_4, v_5, v_6, v_7, v_8$

we have 12 edges { 8 edges 4 chords }

Both with degree 3.

G		H
u_1	u_1u_8, u_1u_2, u_1u_3	v_1
u_2	u_2u_1, u_2u_3, u_2u_6	v_2
u_3	u_3u_3, u_3u_4, u_3u_5	v_3
u_4	u_4u_3, u_4u_5, u_4u_8	v_4
u_5	u_5u_4, u_5u_6, u_5u_3	v_5
u_6	u_6u_5, u_6u_7, u_6u_2	v_6
u_7	u_7u_6, u_7u_8, u_7u_1	v_7
u_8	u_8u_7, u_8u_1, u_8u_4	v_8

simil
-ar

G ↓

Cycles → Min a Δ of 3 is made

H → No Δ

Not Isomorphic.

LIC Aaj Vishvas, Kal Suraksha

2) How many vertices if Degree = 4
4 edges = 10.

→

$n \times \text{degree} = 2 \times \text{edges}$

$\boxed{n = 5}$

3) $K_n \rightarrow$ complete graph 'n' vertices. $n \geq 3$
Let u, v, w be distinct vertices of K_n.

No. of paths distinct from u to v not w?

→

$K_n = K_{n-1}$ → since we omit w.

Path range

$k = 1$ to $k = n-2$ edges.

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All possible path range is
$k = 1$ to $k = n-2$ edges

General $P(k-1, n-3)$

Permutation is

$$\sum_{i=0}^{n-3} = P(n-3, i) = \sum_{i=0}^{n-3} \frac{(n-3)!\, i!}{(n-3-i)!}$$

Simplify

$$P = P_{total} - P_{through\ w}.$$

Ex. if $n = 4$;

$P_{total} = 1 + 2 = 4$ (direct + 1 intermediate)

$P_{through\ w} = 1$ (only $v \to u \to w$)

so

$$\boxed{P = 3}$$

4) → Fig 1. follows Euler path and each edge has even degree so possible vertice.

The points made by intersection create 6 vertices of degree 3 which is odd so

No.

5) Letter

→ Given $m = 5$, $I = 10K$

Total reciepients $n = mI + 1 = 50,001$

People not sending L = $n - I = 50,001 - 10,000$

$= 40,001$

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6) Sort 5

→ $n = 5$ calculating lower bond

$$\log_2 n! = \log_2 120$$
$$= 7$$

Algorithm : Merge Insertion Sort

→ pair 4 compare (2 comparisons)

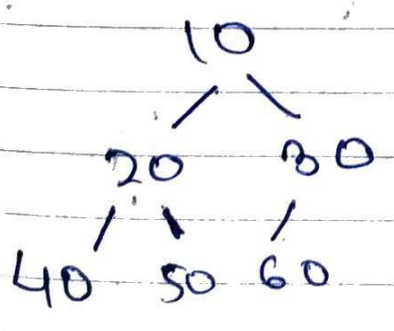
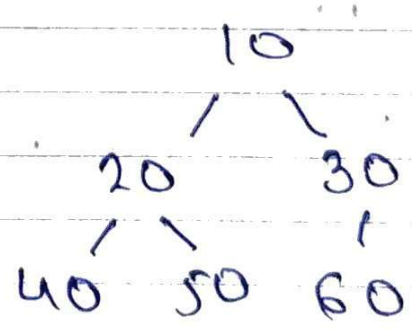
→ Compare winner of pair (1 comparison)

→ Insert remaining using binary insertion (4 comparison)

7) Inorder = [40, 20, 50, 10, 60, 30]

Preorder = [10, 20, 40, 50, 30, 60]

→ Inorder



preorder